

# FAITH OGUNDIMU

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## PROFESSIONAL SUMMARY

Computational cancer-genomics PhD researcher building machine learning for the non-coding genome, with first-author work in minor revisions at npj Genomic Medicine (Nature portfolio). Hands-on deep learning across convolutional, feed-forward and multimodal LLM systems, applied to chromatin accessibility, cfDNA methylation and single-cell multi-omics at the scale of thousands of tumour genomes. Turns large, messy biological data into models, rigorous evaluation and candidates for experimental validation; Government of Ireland Scholar (18% success rate).

## EDUCATION

### Royal College of Surgeons Ireland (RCSI)

Sep 2025 - Aug 2029

Doctor of Philosophy (PhD) in Cancer Genomics and Epigenomics, Machine Learning and Bioinformatics

Thesis: Identification of non-coding somatic driver mutations in breast cancer

### Dublin City University (DCU)

Sep 2021 - May 2025

BSc. in Genetics and Cell Biology (GCB) | First Class Honours (1.1) | Ranked 2nd & Dean's Honour List

Final Dissertation: *Using neural networks and foundation models to understand gene regulation in the MCF-7 breast cancer cell line*

## RESEARCH EXPERIENCE

### Royal College of Surgeons Ireland (RCSI)

Sep 2025 - Aug 2029

#### Doctoral Researcher

- Building ML (Hidden Markov Models, CNNs, MLPs) to prioritise non-coding driver mutations from >5,000 breast-cancer whole genomes (TCGA, ICGC, HMF), with CRISPR-based validation in cell lines.
- Constructing 3D tissue-specific regulatory maps and integrating single-cell ATAC-seq with ENCODE elements to target the 98% non-coding genome where drivers remain uncharacterised.

### Beaumont RCSI Cancer Centre, Irish Cancer Society, Lung Health Check

January 2026 - March 2026

Bioinformatics Research Assistant – BRAND Study (NSCLC Liquid Biopsy & cfDNA Methylation Analysis)

- Built an end-to-end cfDNA methylation and DMR-discovery pipeline across 88 bisulfite-sequenced plasma samples (~8.7M CpGs), reducing >89,000 candidate markers to 100 high-information CpGs (mean  $|\Delta\beta| = 0.57$ ).
- Estimated tumour methylation fraction by semi-reference-free deconvolution for Ireland's largest lung-cancer pilot (€4.9M); found a significant post-immunotherapy TMF drop (Wilcoxon  $p=0.015$ ,  $r=0.85$ ) and  $\Delta$ TMF as a near-significant survival predictor (Cox  $HR=1.14$ ,  $p=0.068$ ).

### University College Dublin

June 2025 - August 2025

Breakthrough Summer Research Scholar (BioSLATE - Biomarker Selection and Synthetic Lethality Analysis for Therapeutic

Exploration in high-grade serous ovarian cancer (HGSOC)) - [BioSLATE Website](#) | [GitHub \(BioSLATE\)](#) | [GitHub \(OncoSynth\)](#)

- Identified synthetic-lethal targets in HGSOC via significant CNA-CRISPR correlations across >500 tumours, integrating TCGA CNA, DepMap CRISPR and gene-essentiality models; shipped an interactive Streamlit app for target nomination.
- Built OncoSynth, a CrewAI multi-agent CLI automating literature mining, druggability scoring and clinical-trial linkage for translational target discovery.

### Royal College of Surgeons Ireland (Final Year Dissertation)

January 2024 - March 2024

Visiting Student Researcher - [GitHub](#)

- Built a feed-forward neural network on the MeluXina HPC to predict MCF-7 chromatin accessibility from nine epigenetic features (>2.8M bins), reaching 98% accuracy and outperforming published models.
- Ranked features with LOCO and SHAP and benchmarked against the GET foundation model, identifying H3K4me1 as the top sequence-independent enhancer predictor.

## PUBLICATIONS

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Ogundimu F. & Furney SJ: Deciphering the Non-coding Cancer Genome through Deep Learning and Multiomic Integration (*in minor revisions - npj Genomic Medicine*)

O'Reilly D, et al.: Biomarkers of Response to Immunotherapy in Advanced Non-small cell lung cancer using Dynamic breath and blood analysis (BRAND) (*in preparation - targeted journal; Journal of Thoracic Oncology*)

Thinnes C. et al.: DrugMap: A genome-scale visualisation of human drug metabolism (*in preparation*)

## GRANTS & SCHOLARSHIPS

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[Government of Ireland \(GOI\) Postgraduate Scholarship](#) (€136,000, 18% success rate) – Nationally competitive award recognising exceptional research potential.

[France Excellence Research Residency Grant](#) (€1,850, French Embassy in Ireland) – Competitively awarded to one of nine awardees nationally; funding a 1-month research residency at Institut Curie (CUBIC, Prof. Nicolas Servant) for CNV-aware chromatin topology analysis and AI-ready regulatory map construction in breast cancer.

**RCSI Top-Up Research Grant (€10,000)** – Selective supplement to GOI Scholarship supporting CRISPR-based validation of computationally predicted non-coding driver mutations in breast cancer.

[Breakthrough Cancer Research Summer Scholarship](#) (€3,000) – National cancer research award supporting 12 hands-on projects in poor-prognosis cancers.

[Unai Fellowship for AI in Biology \(Paris, 2026-27\)](#) – Selected for the inaugural cohort of a competitive year-long fellowship for researchers at the intersection of AI and biology, mentored by leaders in foundation models for biology.

## SELECTED PROJECTS & HACKATHON WINS

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**NCypher — 1st place, Researcher track, Built with Claude: Life Sciences** Jul 2026  
(Anthropic · Cerebral Valley · Gladstone Institutes; 1 of 299)

[GitHub](#) | [Demo Video](#)

- Honest MCP triage tool for non-coding cancer variants, chromatin (ChromBPNet), evolutionary constraint and MPRA function, converged with base-level mechanism.

**Unravel — 1st place, Fivetran track, Google Cloud Rapid Agent Hackathon** Jul 2026

[GitHub](#) | [Live Link](#) | [Demo Video](#)

- Five-agent Gemini 3.1 / Google ADK system that re-scores ACMG pathogenicity when a variant is reclassified and drafts clinician-approved patient recontact (FHIR).

## CONFERENCES, PRESENTATIONS & TRAINING

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**Invited speaker, Anthropic Claude Science AMA (Aug 2026)** - live demo and Q&A on deploying AI agents in cancer epigenomics research, alongside Claude Science lead engineer, Alexander Tarashansky.

**Imperial College London Robotics Summer School (Jul 2026)** - Hamlyn Centre for Robotic Surgery.

**Invited panel speaker, UCD Women+ in STEM Conference, sponsored by AMD (Apr 2026)** - STEM careers, academia vs industry, the future of computational biology.

**Posters:** Conway Institute Research Festival, UCD (2025); RCSI Research Day (2026); BRS Research Day, DCU (2025).

## ACADEMIC DISTINCTIONS & AWARDS

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**Conway Institute Research Festival** - Poster Commendation (perfect 30/30).

**Breakthrough Cancer Research** - Best Lay Presentation Award (2025).

**BSc. Genetics & Cell Biology** - Faculty Undergraduate Academic Achievement Award; FSH Academic Scholarship.

## SKILLS

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**Programming:** Python, R, Bash, Perl, TypeScript

**Machine Learning & AI:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, deep learning (CNNs, MLPs), multi-agent systems (CrewAI, Google ADK), RAG, LangChain

**Genomics & Data Science:** NGS, single-cell ATAC-seq, cfDNA/methylation & bisulfite analysis, survival analysis (Kaplan–Meier, Cox), NMF/NNLS deconvolution, TCGA/DepMap/ENCODE, Docker

**Cloud & App:** Google Cloud Platform, Vertex AI, FastAPI, Streamlit, React.js, Firebase